

JAVA LAB RECORD

Experiment No: 06

Date: 15/09/2026

Title: Exception Handling using Try-Catch in Java

■ Objective / Aim

To demonstrate exception handling in Java using try-catch with `ArrayIndexOutOfBoundsException`, `ArrayStoreException`, `IllegalArgumentException`, `NegativeArraySizeException`, `NullPointerException`, `NumberFormatException`, and `StringIndexOutOfBoundsException`.

■■ System Requirements

Java Development Kit (JDK), any Java IDE or command prompt, and a system capable of compiling and executing Java programs.

■■ Algorithm / Logic Description

1. **ArrayExample:** Create an integer array, access an element inside a try block, and handle an invalid index using `ArrayIndexOutOfBoundsException`.
2. **ArrayStoreExample:** Create a String array, refer to it through `Object[]`, attempt to store an integer, and catch `ArrayStoreException`.
3. **IllegalArgumentExceptionExample:** Define `setAge()`, check whether age is negative, explicitly throw `IllegalArgumentException` when it is negative, and catch it in `main()`.
4. **NegativeArrayExample:** Set the array size to -5, attempt to create the array inside a try block, and catch `NegativeArraySizeException`.
5. **NullPointerExceptionExample:** Assign null to a String variable, attempt to call `length()` on it, and catch `NullPointerException`.
6. **NumberFormatExceptionExample:** Store a non-numeric String, attempt to convert it to an integer using `Integer.parseInt()`, and catch `NumberFormatException`.

7. **StringExample:** Store a String, attempt to access character index 7 even though the word has only five characters, and catch `StringIndexOutOfBoundsException`.

■ UML Class Diagram

```
ArrayExample
|
+-- main()

ArrayStoreExample
|
+-- main()

A
|
+-- setAge(int)

IllegalArgumentExample
|
+-- main()
|
+-- uses A

NegativeArrayExample
|
+-- main()

NullPointerExample
|
+-- main()

NumberFormatExample
|
+-- main()

StringExample
|
+-- main()
```

■ Source Code – ArrayExample.java

```
public class ArrayExample {
```

```

public static void main(String[] args) {
    int number[] = { 10, 20, 30 };
    try {
        System.out.println(number[1]);
    } catch (ArrayIndexOutOfBoundsException aioobe) {
        System.out.println("index does not exist");
    }
}
}

```

■ Source Code – ArrayStoreExample.java

```

public class ArrayStoreExample {
    public static void main(String[] args) {
        try {
            String[] Student=new String[3];
            Object[] o=Student;
            o[0]=579;
            o[1]="hi";
            o[2]="hello";
            for (int i = 0; i < 3; i++) {
                System.out.println(o[i]);
            }
        } catch(ArrayStoreException ase) {
            System.out.println("array must contain Strings");
        }
    }
}

```

■ Source Code – IllegalArgumentExample.java

```

class A{
    void setAge(int age){
        if(age<0){
            throw new IllegalArgumentException("cannot be negative");
        }
        System.out.println("age is:"+age);
    }
}

public class IllegalArgumentExample{
    public static void main(String[] args) {
        A a=new A();
        try{
            a.setAge(-5);
        } catch(IllegalArgumentException iae){
            System.out.println(iae.getMessage());
        }
    }
}

```

```
}
```

```
}
```

■ Source Code – NegativeArrayExample.java

```
public class NegativeArrayExample {
    public static void main(String[] args) {
        int i = -5;
        try {
            int[] numbers = new int[i];
            numbers[0] = 10;
            numbers[1] = 20;
            numbers[2] = 30;
            numbers[3] = 40;
            numbers[4] = 50;
            System.out.println(numbers[2]);
            System.out.println("array is created");
        } catch (NegativeArraySizeException na) {
            System.out.println("array size must be positive");
        }
    }
}
```

```
}
```

■ Source Code – NullPointerException.java

```
public class NullPointerException {
    public static void main(String[] args) {
        String name = null;
        try {
            System.out.println(name.length());
        } catch (NullPointerException e) {
            System.out.println("it cannot perform length() because name contain null means empty");
        }
    }
}
```

```
}
```

■ Source Code – NumberFormatException.java

```
public class NumberFormatException {

    public static void main(String[] args) {
        String value = "hellloo";
        try {
            int num = Integer.parseInt(value);
            System.out.println("String converted to number=" + num);
        } catch (NumberFormatException nfe) {
            System.out.println("unable to convert ");
        }
    }
}
```

```
    }  
}  
  
}
```

■ Source Code – StringExample.java

```
public class StringExample {  
    public static void main(String[] args) {  
        String word = "hello";  
        try {  
            System.out.println(word.charAt(7));  
        } catch (StringIndexOutOfBoundsException sioobe) {  
            System.out.println("Index does not exist");  
        }  
    }  
}
```

■ Compilation & Execution Commands

```
javac  
ArrayExamp  
e.java java  
ArrayExamp  
e
```

```
javac  
ArrayStoreExa  
mple.java  
java  
ArrayStoreExa  
mple
```

```
javac  
IllegalArgument  
Example.java  
java  
IllegalArgument  
Example
```

```
javac  
NegativeArrayE  
xample.java  
java
```

```
NegativeArrayE  
xample
```

```
javac  
NullPointerException  
ample.java  
java  
NullPointerException  
ample
```

```
javac  
NumberFormatEx  
ample.java  
java  
NumberFormatEx  
ample
```

```
javac  
StringExamp  
le.java  
java  
StringExamp  
le
```

■■ Output Screen

```
C:\Java>javac  
ArrayExample.java  
C:\Java>java ArrayExample  
20  
C:\Java>  
  
C:\Java>javac  
ArrayStoreExamp  
le.java  
C:\Java>java  
ArrayStoreExamp  
le array must  
contain Strings  
C:\Java>
```

```
C:\Java>javac  
IllegalArgumentEx  
ample.java
```

```
C:\Java>java  
IllegalArgumentEx  
ample cannot be  
negative C:\Java>
```

```
C:\Java>javac  
NegativeArrayExample.java
```

```
C:\Java>java  
NegativeArrayE  
xample array  
size must be  
positive  
C:\Java>
```

```
C:\Java>javac  
NullPointerException.j  
ava C:\Java>java  
NullPointerException  
it cannot perform length()  
because name contain null  
means empty C:\Java>
```

```
C:\Java>javac  
NumberFormatExa  
mple.java  
C:\Java>java  
NumberFormatExa  
mple unable to  
convert  
C:\Java>
```

```
C:\Java>javac  
StringExample.java  
C:\Java>java StringExample  
Index does not exist  
C:\Java>
```

■ Concepts Demonstrated

- try-catch exception handling
- `ArrayIndexOutOfBoundsException`
- `ArrayStoreException`
- `IllegalArgumentException`
- `NegativeArraySizeException`
- `NullPointerException`
- `NumberFormatException`
- `StringIndexOutOfBoundsException`
- throw statement
- `Integer.parseInt()`
- `getMessage()` method
- Runtime exception handling

■ Result

The Java programs were successfully designed to demonstrate handling of different runtime exceptions using try-catch and explicit exception throwing.